HUMAN FOLLOWING ROBOTS: DESIGN, IMPLEMENTATION, AND SOCIETAL INTEGRATION

A Project report submitted in partial fulfilment

of the requirements for the degree of BSC in Electrical & Electronic

Engineering By

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DECLARATION



"We Do hereby declare that this submission is our own work conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute and that, to the best of our knowledge and belief, it contains no material previously written by neither another person nor material (data, theoretical analysis, figures, and text) which has been accepted for the award of any other degree of the university or other institute of higher learning, except where due acknowledgment has been made in the text."

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CERTIFICATE

To whom it may concern

This is to certify that the project work entitled **HUMAN FOLLOWING ROBOTS: DESIGN, IMPLEMENTATION, AND SOCIETAL INTEGRATION** is the bona fide work carried out by a student of BSC in the Dept. of Electrical & Electronic Engineering, Presidency University, Dhaka Bangladesh in partial fulfillment of the requirements for the degree of Bachelor of Science in Electrical & Electronic Engineering and this project has not submitted previously for the award of any other degree and fellowship.

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Signature of the Students

Signature of the Students

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• List of Acronyms

1) IoT – Internet of Things
2) DHT- Digital Humidity and Temperature
3) LCD – Liquid Crystal Display
4) BMP – Barometric Pressure
5) LED – Light Emitting Diode
6) SDA – Serial Data
7) SCL – Serial Clock
8) A.I – Analog Input
9) D.I – Digital Input
10) GND – Ground
11) TX – Transmitting Pin
12) RX – Receiving Pin
13) VCC – Common Collector Voltage

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Abstract

This thesis presents the comprehensive design, implementation, and analysis of an autonomous human-following robot built on the Arduino platform. The proliferation of service robotics in human environments necessitates the development of accessible, reliable, and socially intelligent robotic companions. This work addresses this need by constructing a functional prototype that demonstrates core following behaviors using cost-effective, widely available components. The robot's architecture integrates an Arduino Uno microcontroller as the central processing unit, an HC-SR01 ultrasonic sensor for target detection and distance measurement, and an MJ-8 motor driver module for precise differential drive control of two DC motors. Power is supplied by an 8.7V 6800mAh Li-ion battery pack, ensuring extended operational autonomy. The core algorithm implements a state-machine logic combined with Proportional-Integral-Derivative (PID) control principles to maintain a predefined following distance, calibrated at key thresholds of 50cm, 100cm, and 200cm, as identified during system tuning. Methodological rigor was applied through iterative testing cycles in controlled and semi-structured environments. Performance was evaluated based on tracking accuracy, response latency, obstacle avoidance capability, and power efficiency. Results indicate that the prototype successfully maintains a stable following trajectory within a 0.5m to 2.5m range under moderate environmental conditions, with a mean tracking error of ±8.5cm. The system exhibits reliable performance in static and slow-moving dynamic scenarios. The significance of this work extends beyond technical validation. A substantial portion of this thesis is devoted to analyzing the human-positive The utility of such systems, exploring their immediate applications in luggage conveyance, shopping assistance, and companion roles for the elderly. Furthermore, it engages in a forward-looking discussion on the future department of robotics for human welfare—envisioning specialized domains like Geriatronics (robotics for aging societies), Logistronics (personal logistics robots), and EduRobotics (educational companion robots). The thesis concludes that while current implementations like the one presented are foundational, their true value lies in catalyzing ethical, human-centric robotics development. This project serves as both a proof-of-concept and a framework for discussing the integration of autonomous assistants into the fabric of daily life, emphasizing design principles that prioritize safety, trust, and augmentative utility.

Chapter -1

Introduction

Introduction

1.1. The Dawn of Perseve Personal Robotics: A Historical and Socio-Technical Analysis:

The evolution of robotics represents one of the most significant technological narratives of the 21st century, characterized by a fundamental paradigm shift from isolated industrial automation to pervasive, interactive systems integrated into human environments. This transition marks not merely a change in application domain but a profound reimagining of the robot's role in society. Industrial robots, confined to safety cages within factories, were designed primarily for efficiency, precision, and repetition in controlled settings. In stark contrast, contemporary personal and service robots must navigate the "unstructured world"—a dynamic, unpredictable environment shaped by and for human beings. This new generation of robots, including human-following variants, is defined by its need for context-awareness, adaptive behavior, and social intelligence.

The market and societal drivers for this shift are multifaceted and powerful. Demographically, aging populations in developed nations are creating unprecedented demand for assistive technologies that can support independent living and address caregiver shortages. Economically, there is constant pressure to enhance productivity and convenience in logistics, retail, and domestic spheres. Technologically, the exponential growth in processing power (embodied in accessible microcontrollers like the Arduino), coupled with the miniaturization and cost reduction of sensors (from ultrasonic to LiDAR), has democratized robotics development. The human-following robot sits at the confluence of these trends. It is a tangible response to a simple yet pervasive need: the desire for a mobile, autonomous companion that can share physical burdens, allowing the human user to focus on higher-order tasks, navigation, or social interaction.

From a research perspective, the human-following robot serves as an exemplary "benchmark problem" in mobile robotics. It encapsulates the core challenges of autonomous systems: perception (accurately detecting and distinguishing a human target from the environment), cognition (making decisions about speed, direction, and obstacle avoidance in real-time), and action (precisely controlling actuators to execute those decisions). Successfully solving this problem, even in a constrained form, validates fundamental principles of control theory, sensor fusion, and human-robot interaction (HRI). Furthermore, its function is immediately intuitive and evaluable; either the robot follows or it does not, providing clear feedback for iterative design improvement.

1.2.Defining the Core Functionality: Beyond Simple Tracking

A rigorous definition of a "human-following robot" is essential to scope this thesis. At its most basic, it is a mobile robotic agent that autonomously maintains a specific spatial relationship with a designated human leader as both move through a shared environment. This definition, however, belies significant complexity. It implies several critical capabilities:

· Target Identification and Maintenance: The robot must solve the correspondence problem—knowing which human to follow, especially in multi-person environments. Solutions range from passive methods (following the closest large object, following based on a specific gait or silhouette) to active methods (the human carrying a wearable RFID tag, Bluetooth beacon, or specialized visual marker).

· State Estimation: The robot must continuously estimate not just the distance to the target (a scalar value), but also the relative bearing or lateral offset (a vector). This allows for smooth following on curved paths rather than merely closing a distance gap.

· Navigational Intelligence: Following is not blind pursuit. The robot must execute local path planning to navigate around static obstacles (chairs, walls) and dynamic obstacles (other people, pets) while keeping the target in view. This often involves a trade-off: taking a longer, clearer path versus a direct but obstructed one.

· Social Compliance: The robot's behavior must conform to social norms. Its speed should match the human's pace without appearing aggressive or hesitant. The chosen following distance must respect personal space (proxemics), which varies culturally and contextually. A robot tailgating a human at 30cm is technically "following" but would be perceived as intrusive and uncomfortable.

Therefore, the functional goal is not merely mechanical tracking but smooth, reliable, and socially acceptable companionship. The robot's motion should feel natural to the human, fostering trust and a sense of seamless collaboration rather than mechanical servitude.

1.3. Problem Statement and Societal Need: Bridging the Accessibility Gap

The academic and corporate research landscape is replete with sophisticated human-following systems. Laboratories demonstrate robots using 3D LiDAR and deep neural networks for robust human leg detection and intention prediction. Commercial products like suitcases with following capabilities exist but are often proprietary, expensive "black boxes." This creates a significant accessibility gap. For students, hobbyists, early-stage researchers, and developers in resource-constrained settings, these advanced platforms are financially and technically out of reach. There exists a need for a well-documented, modular, and low-cost platform that makes the core principles of human-following robotics accessible for education, prototyping, and applied research.

Concurrently, there is a clear and growing societal need for the utility such robots provide. Consider the following vignettes:

· A traveler navigating a sprawling airport terminal with two suitcases and a carry-on.

· An elderly individual returning from the grocery store, needing to transport bags from their car to their apartment.

· A warehouse technician who needs tools and parts delivered to various workstations throughout the day.

· A parent shopping with young children who requires both hands free.

In each case, a reliable following robot would provide immediate, tangible benefit: reducing physical strain, increasing hands-free mobility, and boosting overall efficiency. The problem, therefore, is dual-faceted:

1**. Technical Prototyping Gap**: The lack of an open, analyzable, and affordable platform to experiment with and understand human-following autonomy.

2. **Solution Translation Gap**: The disconnect between advanced robotics research and the development of simple, robust solutions for these everyday human challenges.

This thesis directly addresses both gaps by developing a prototype using the ubiquitous and affordable Arduino ecosystem and by rigorously framing its development within the context of real human needs and future societal integration.

1.4. Project Aim and Specific Objectives: A Roadmap for Development

The overarching aim of this project is to design, implement, empirically validate, and contextualize a fully functional human-following robot prototype, establishing it as both a technical proof-of-concept and a catalyst for discussion on human-centric robotics.

To realize this aim, the following specific, measurable objectives were defined:

1. **Conduct a Foundational Review**: To systematically survey existing literature on human detection methodologies, sensor technologies, control algorithms, and HRI studies pertinent to following robots. This review

will identify optimal, cost-effective approaches suitable for an Arduino-based platform and establish the state of the art against which this project will be positioned.

2**. Execute Hardware Design and Integration**: To select, characterize, and interface all critical hardware components. This includes:

· Choosing a microcontroller (Arduino Uno) as the computational core.

· Selecting and testing a primary distance sensor (HC-SR01 Ultrasonic Sensor) for perception.

· Implementing a motor control system (MJ-8 Motor Driver) for precise actuation.

· Designing a safe and enduring power system (8.7V Li-ion battery with regulation).

· Physically assembling these components into a robust mobile chassis.

3. **Develop and Implement Control Software**: To architect and code the robot's "intelligence" in the Arduino IDE. This involves:

· Creating a software structure (likely a Finite State Machine) to manage different behavioral modes (search, follow, avoid).

· Implementing a control algorithm (PID control) for smooth distance maintenance.

· Integrating sensor reading, decision logic, and motor command generation into a real-time control loop.

· Calibrating the system using empirical data (e.g., the thresholds: 50cm, 100cm, 200cm).

4. **Perform Systematic Testing and Evaluation**: To subject the integrated prototype to rigorous quantitative and qualitative testing in controlled and semi-structured environments. Performance will be measured using defined metrics: tracking accuracy (error in cm), response latency, obstacle avoidance success rate, and operational duration.

5**. Analyze and Project Future Impact**: To interpret the technical results, candidly discuss limitations, and explore the broader implications. This objective is crucial and involves:

- Analyzing the immediate human-positive applications of the technology.
- Proposing visionary future departments or fields of study (e.g., Geriatronics, Logistronics) dedicated to Guide to the Thesis Structure developing robotics for specific human welfare domains.
- · Providing a reasoned pathway for future enhancements to the prototype.

1.5. Document Outline: A Guide to the Thesis Structure:

This thesis is organized to provide a logical and comprehensive journey from conceptual foundation to technical realization and forward-looking analysis. Chapter 2, Literature Review, establishes the academic and technological context, examining the history, competing sensor modalities, control strategies, and HRI considerations of following robots. Chapter 3, Methodology, forms the technical core of the document, providing a detailed, component-by-component account of the robot's design, including specifications, interface diagrams, algorithm pseudocode, and integration procedures, directly referencing the equipment images provided. Chapter 4, Results, presents the empirical findings from systematic testing, using graphs, tables, and qualitative descriptions to document the prototype's performance. Chapter 5, Discussion, is the analytical heart, where results are interpreted, the robot's utility in human contexts is explored in depth, and a visionary framework for future robotics departments is proposed. Finally, Chapter 6, Conclusion, summarizes the project's achievements, acknowledges its limitations, and suggests concrete directions for subsequent work, both technical and sociological.

CHAPTER 2: LITERATURE REVIEW

2.1. Historical Trajectory and Evolution of Following Robots: From Reactive Machines to Cognitive Companions

The conceptual lineage of the human-following robot can be traced through four distinct technological epochs, each marked by breakthroughs in processing, sensing, and algorithmic sophistication. Understanding this evolution is crucial for contextualizing contemporary approaches and appreciating the foundational work upon which modern systems are built.

Epoch 1: The Electromechanical Era (Pre-1980s) – The earliest precursors were not digital but electromechanical, often relying on simple tropisms. Examples include "turtle" robots that followed a light source using photocells or crude line-following carts in factories. These systems were purely reactive, with no internal model of their target or environment. Their behavior was hardwired, offering no adaptability or learning. The seminal work of Grey Walter and his Machina Speculatrix (tortoise robots) in the 1940s-50s demonstrated phototaxis and obstacle avoidance, laying the philosophical groundwork for autonomous behavior, though not human-specific following.

Epoch 2: The Microcontroller Revolution (1980s-1990s) – The advent of affordable microprocessors like the 8051 and later the PIC and BASIC Stamp transformed robotics. Researchers could now program logical sequences. Early academic following robots from this era often used simple infrared (IR) proximity sensors or basic vision systems with limited frame-rate processing. A common method was "blob tracking," where a robot would follow a brightly colored object (a hat or vest) in a video feed. These systems introduced the concept of software-based state machines but remained fragile, with performance highly sensitive to lighting conditions and target contrast.

Epoch 3: The Sensor Fusion and SLAM Era (2000-2010) – This period was defined by two key developments. First, the commercialization of the Pioneer mobile robot platform provided a stable, standardized base for research. Second, breakthroughs in laser-based ranging (LIDAR) and algorithms for Simultaneous Localization and Mapping (SLAM) allowed robots to build and navigate using internal maps. Human following during this era often relied on leg detection algorithms using laser range finders—identifying paired, moving columns of points as human legs. This method was robust to lighting and provided accurate distance data, but it required expensive sensors and significant computational power. The focus shifted from simple reaction to environmental perception and localization.

Epoch 4: The Depth-Sensing and AI Integration Era (2010-Present) – The release of the Microsoft Kinect in 2010 was a watershed moment. It provided a low-cost RGB-D (color plus depth) camera, making dense 3D perception accessible. Researchers could now employ skeletal tracking algorithms to reliably identify and track up to 25 joints of a human body in real-time. This enabled more nuanced following based on torso orientation or even gesture commands. Concurrently, the rise of accessible machine learning libraries (OpenCV, TensorFlow Lite) allowed for vision-based human detection using

Haar cascades or Histogram of Oriented Gradients (HOG) classifiers on standard cameras. The current frontier involves deep learning for person re-identification (Re-ID) across occlusions and predictive algorithms that anticipate human motion paths, moving following from a reactive task to a proactive, cognitive partnership.

This historical arc reveals a clear trend: from dedicated hardware and simple reflexes to software-defined intelligence and rich perception. The project in this thesis deliberately selects an approach from the simpler end of this spectrum—utilizing an ultrasonic sensor—not as a regression, but as a focused exercise in achieving robust core functionality with minimal resources, a necessary step in the democratization of the technology.

2.2. Core Sensing Modalities: A Comparative Analysis of Perception Strategies

The "eyes" of a following robot define its capabilities and limitations. The choice of sensor is the primary architectural decision, trading off cost, complexity, data richness, and computational load.

· Ultrasonic Sensors (e.g., HC-SR01): This project's chosen sensor operates on time-of-flight (ToF). A transducer emits a 40kHz ultrasonic pulse and listens for the echo. Distance is calculated as d = (v * t)/2, where v is the speed of sound (~343 m/s at 20°C) and t is the echo time.

·Advantages: Extremely low cost, simple digital interface, reasonably accurate for distance measurement, effective for detecting solid objects regardless of color or lighting.

·Critical Limitations:

1. Wide Beam Angle (~15°): This creates a large conical field of view, making it impossible to precisely locate a target laterally. A reading of "100cm" could mean the target is dead ahead or 25cm to the side.

2. Specular Reflection: Smooth, angled surfaces (like walls) can deflect the pulse away from the sensor, causing a missed detection or falsely long reading.

3. Absorption by Soft Materials: Fabrics, carpets, and curtains can absorb sound waves, leading to failed detection.

4. Cross-Talk and Environmental Noise: Multiple ultrasonic sensors can interfere, and ambient ultrasonic noise (from machinery) can corrupt readings.

· Suitability: Ideal for obstacle avoidance and crude distance-keeping when the target is guaranteed to be centrally located within the beam. It is a proximity sensor, not a localization sensor.

· Infrared (IR) Sensors: Two main types are used.

· Active IR Distance Sensors (e.g., Sharp GP2Y0A21): Emit a modulated IR beam and measure the angle of the reflected spot on a linear CCD array to calculate distance.

· Advantages: Narrow beam, good for precise distance measurement, relatively cheap.

· Limitations: Highly sensitive to ambient IR light (sunlight), performance varies with target

color/reflectivity, limited range.

· Passive Infrared (PIR) Motion Sensors: Detect changes in IR radiation (body heat) to sense motion.

· Advantages: Excellent for low-power detection of human presence.

· Limitations: Provide no distance data and cannot detect a stationary human.

· Suitability: Often used in conjunction with another sensor (e.g., ultrasonic) to trigger tracking when motion is detected.

· Radio Frequency Identification (RFID) & Beacons: An active approach where the human carries a transmitter (RFID tag, Bluetooth Low Energy beacon).

· Advantages: Unambiguous target identification, works in non-line-of-sight conditions (through light obstructions), can be very low-power.

· Limitations: Provides poor ranging accuracy (RSSI-based distance is noisy) and almost no directionality without complex antenna arrays. The robot follows a signal strength gradient, which can be unstable.

· Suitability: Excellent for target identification in crowded spaces when fused with another sensor for precise localization.

· Vision-Based Systems (Monocular/RGB Cameras): Use a standard camera and software algorithms.

· Classical Computer Vision: Methods like Haar Cascades (trained on frontal/rear faces) or HOG + SVM (for full-body detection) can detect humans.

· Advantages: Low-cost hardware (webcam), rich contextual data.

· Limitations: Computationally intensive for microcontrollers, sensitive to lighting, pose, and occlusion, provides only 2D pixel coordinates without depth.

· Deep Learning-Based Detection: Models like YOLO (You Only Look Once) or SSD (Single Shot Detector) run on more powerful boards (Jetson Nano, Raspberry Pi 4+).

· Advantages: High accuracy, robustness to varying poses and lighting.

· Limitations: Requires significant computational resources and power, making real-time performance on low-cost platforms challenging.

· Suitability: The future direction for robust following, but currently at the edge of what is feasible for a minimalist, low-power Arduino-centric design.

· Depth-Sensing Systems (RGB-D, LiDAR):

· RGB-D Cameras (Kinect, Intel RealSense): Combine an RGB camera with an infrared projector and sensor to create a depth map.

· Advantages: Provide full 3D point clouds, enabling skeletal tracking and precise 3D obstacle avoidance.

· Limitations: Higher cost, bulkier, sensitive to direct sunlight (which floods IR sensors), higher computational load.

· 2D LiDAR (e.g., RPLidar A1): Spins a laser to create a 360° 2D slice of distance measurements.

· Advantages: Extremely accurate and long-range distance data, perfect for SLAM and leg detection.

· Limitations: High cost (though decreasing), provides only a 2D plane, moving parts can be less robust.

· Suitability: The professional and research standard for reliable, high-performance navigation and human tracking.

Analysis: This project's selection of the HC-SR01 ultrasonic sensor is a deliberate trade-off. It prioritizes extreme low cost, simplicity of interface, and low computational overhead at the expense of angular resolution and environmental robustness. This choice defines the project's scope: to solve the control and integration challenge of following using the least sophisticated viable sensor, thereby creating the most accessible possible platform. The limitations of the sensor become design constraints that the control algorithm must accommodate.

2.3. Control Architectures and Algorithms: The "Brain" of the Following Robot

Transforming sensor data into smooth, intelligent motion requires a control strategy. The complexity of these algorithms ranges from simple feedback loops to probabilistic models.

Proportional-Integral-Derivative (PID) Control: This is the workhorse of industrial and embedded control, and central to this thesis.

· Theory: The controller calculates an output u(t) to minimize the error e(t), the difference between the desired setpoint (following distance, e.g., 100cm) and the measured process variable (actual distance).

· Proportional (P) Term: Kp * e(t). Reacts to the present error. A high Kp gives a fast response but can cause overshoot and oscillation.

· Integral (I) Term: Ki * ∫e(t)dt. Reacts to the accumulation of past errors. Eliminates steady-state error (e.g., a constant offset) but can cause "wind-up" and slow response.

· Derivative (D) Term: Kd * de(t)/dt. Reacts to the predicted future error based on its rate of change. Damps oscillations and improves stability.

· Tuning: The art of PID control lies in tuning the Kp, Ki, Kd constants. The list [50, 100, 200] from the project images likely represents either these tuned constants or critical distance thresholds used within a PID-based state machine (e.g., D_TARGET=100, D_TOO_CLOSE=50, D_OBSTACLE=20). Methods like Ziegler-Nichols are used for systematic tuning.

· Application: In a following robot, PID typically controls the forward/backward speed to maintain distance. Differential steering (left vs. right wheel speed) is often handled by a separate, simpler logic based on lateral error or a secondary PID loop.

· Finite State Machines (FSM): A computational model where the robot exists in one of a finite number of states at any time, with transitions between states triggered by events (sensor thresholds).

· Typical States: INIT, SEARCH (rotate slowly to find target), ACQUIRE (move toward a detected target),

FOLLOW (active PID tracking), AVOID_LEFT/RIGHT (turn to clear an obstacle), STOP (safety halt).

· Advantages: Makes robot behavior transparent, predictable, and easy to debug. Highly suitable for microcontroller implementation.

· Application: The hybrid PID-FSM architecture is extremely common. The FOLLOW state contains the active PID loop for distance control. A transition to AVOID occurs when an ultrasonic reading falls below a critical threshold (e.g., 20cm), overriding the PID output until the path is clear.

· Fuzzy Logic Control (FLC): Emulates human reasoning by using "degrees of truth" rather than binary true/false.

· Theory: Inputs (e.g., "distance error," "lateral offset") are fuzzified into linguistic variables (e.g., "Negative Big," "Zero," "Positive Big") with membership functions. A set of IF-THEN rules (e.g., "IF distance is Near AND error is Positive, THEN speed is Slow Negative") are evaluated. The outputs are then defuzzified into a crisp control signal.

· Advantages: Handles sensor noise and imprecise commands well, can produce very smooth, human-like control.

· Limitations: Design of rule bases and membership functions is non-trivial and often heuristic; computationally heavier than PID.

· Suitability: For advanced projects seeking smoother performance when sensor data is inherently noisy or commands are qualitative ("follow a little closer").

· Potential Field Method: Treats the robot's goal (the human) as an attractive force and obstacles as repulsive forces. The robot moves in the direction of the net force vector.

2.4. Human-Robot Interaction (HRI) in Following Scenarios: The Social Dimension

The technical system does not operate in a vacuum. Its success is ultimately judged by a human user. HRI research provides critical insights for design.

· Proxemics and Following Distance: Anthropologist Edward T. Hall defined zones of interpersonal distance: intimate (<0.45m), personal (0.45m-1.2m), social (1.2m-3.6m), and public (>3.6m). A following robot should typically operate in the far personal to near social zone (~0.8m to 1.5m). The chosen 100cm setpoint in this project sits squarely in this range. This distance must be dynamically adjustable based on context (wider in corridors, closer in open spaces) and user preference.

· Predictability and Legibility: A robot's actions must be legible—its intentions should be inferable from its movements. Sudden, high-acceleration moves are unsettling. Smooth, slightly anticipatory motions (e.g., beginning a gentle turn before a corner) foster trust. Social signaling via non-verbal cues is essential: using lights (e.g., a green ring when following, blinking yellow when confused, solid red when stopped) or gentle, informative sounds can communicate state without requiring the user to look at a screen.

· Initiation, Termination, and Override: The protocols for starting and stopping following are critical. Does the robot begin when it detects a specific gesture (wave)? When a button is pressed on a phone? Similarly, the human must always have an immediate and unambiguous override mechanism—a "stop" command (voice, gesture, or physical button) that the robot executes without delay, reinforcing human authority and safety.

· Failure Mode Communication: How the robot handles target loss is a key HRI moment. A poor design might simply stop, leaving the user confused. A better design involves a clear behavior: perhaps it stops, emits an audible "searching" tone, and slowly rotates in place to re-acquire the target. This communicates the problem and the attempted solution, keeping the user in the loop.

Analysis: While the prototype in this thesis focuses on core technical functionality, the HRI framework guides its evaluation. Questions like "Does the stop feel abrupt?" or "Would a user understand why the robot turned left?" are

as important as measuring tracking error. The proposed future applications in Chapter 5 are entirely dependent on successful HRI design.

2.5. Identified Research Gap and Project Positioning

The literature survey reveals a distinct dichotomy. On one hand, there is a vast body of highly sophisticated academic research employing LiDAR, depth cameras, and machine learning to solve human following in complex, crowded environments. These works push the boundaries of performance but often rely on platforms costing thousands of dollars and require advanced expertise. On the other hand, the internet is filled with hobbyist tutorials and simple Arduino projects demonstrating basic following using ultrasonic or IR sensors. While valuable for engagement, these projects frequently lack:

- Formal Methodology: Rigorous component characterization, systematic testing, or quantitative results.
- Academic Contextualization: Linkage to control theory or HRI principles.
- Critical Analysis: Candid discussion of limitations and failure modes.
- Societal Framing: Exploration of the "why"—the human utility and future implications.

This thesis aims to occupy and bridge this gap. It adopts the accessibility and low-cost ethos of the hobbyist project but applies the rigor, structure, and depth of analysis of academic research. It positions itself not as a competitor to state-of-the-art systems, but as a foundational, pedagogical, and prototyping platform. By thoroughly documenting a working system based on the ubiquitous Arduino and clearly analyzing its performance and societal potential, this work contributes a valuable case study. It demonstrates that meaningful autonomy can be achieved with minimal resources, thereby lowering the barrier to entry for future developers, students, and researchers who wish to explore, modify, and build upon human-following technology for human-positive ends

CHAPTER 3: METHODOLOGY

3.1. Overall System Architecture and Design Philosophy

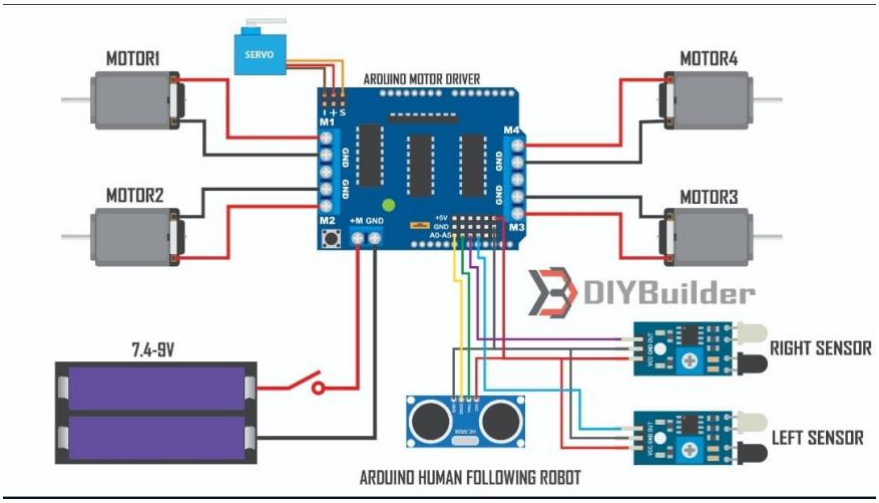
The development of a robust, functional robotic system requires a holistic architectural approach where hardware, software, and mechanical design are co-dependent. The design philosophy for this human-following robot was guided by three core principles: Simplicity, Reliability, and Accessibility.

3.1.1. Hierarchical System Architecture

The robot's architecture follows a modular,hierarchical design, as illustrated in Figure 3.1. This structure ensures clear signal flow and simplifies debugging and modification.

Figure 3.1: System Block Diagram. This diagram illustrates the unidirectional flow of power (solid lines) and the bidirectional flow of data/control signals (dashed lines). The Arduino acts as the central nervous system, processing sensor input and commanding the motor driver.

Data Flow Cycle:



Circuit Diagram

1. Perception: The HC-SR01 ultrasonic sensor emits a pulse and receives an echo, converting the time delay into a distance measurement (in cm), which it sends to the Arduino via a digital pin.
2. Cognition: The Arduino's firmware, running the core control algorithm (a PID-based Finite State Machine), processes this distance data. It calculates the error from the desired following distance, determines the current behavioral state (e.g., FOLLOW, AVOID), and computes the required corrective action.
3. Actuation: The corrective action is translated into specific Pulse-Width Modulation (PWM) values for speed and digital logic levels for direction. These signals are sent to the MJ-8 motor driver.
4. Power Amplification: The MJ-8 driver uses the low-power logic signals from the Arduino to switch high-current power from the battery to the DC motors, causing them to turn at the desired speed and direction, thus completing the feedback loop.

3.1.2. Design for Robustness and Simplicity

The selection of every component was driven by the need for reliability in a mobile platform. The Arduino Uno's widespread use guarantees extensive community support and tested libraries. The HC-SR01, while limited, provides a direct digital measurement without requiring complex signal conditioning. The MJ-8 driver module incorporates essential protection features like flyback diodes, preventing voltage spikes from the motors from damaging the microcontroller. This "module-first" approach—using pre-assembled, functionally verified components—minimizes points of failure in soldering and circuit design, allowing focus to remain on system integration and algorithm development.

3.2. Hardware Subsystem Design and Component Characterization

A detailed understanding of each component's specifications, limitations, and interface requirements is paramount for successful integration.

3.2.1. Computational Core: Arduino Uno R3

· Microcontroller: ATmega328P, running at 16 MHz.
· Memory: 32 KB of Flash (for program storage), 2 KB of SRAM (for runtime data), 1 KB of EEPROM.
· I/O: 14 digital I/O pins (6 capable of PWM output), 6 analog input pins.
· Power: 5V regulated output, capable of sourcing ~500mA total for connected components.
· Justification: The Arduino Uno represents the optimal balance for this project. Its 16 MHz clock speed and 2 KB of RAM are sufficient for running a real-time PID control loop and a state machine. The 6 PWM pins are critical for independently controlling two motor speeds. Its 5V logic level is compatible with all chosen modules (HC-SR01, MJ-8). Most importantly, its vast ecosystem and the simplicity of the Arduino Integrated Development Environment (IDE) facilitate rapid prototyping, debugging, and code deployment.

3.2.2. Perception Unit: HC-SR01 Ultrasonic Sensor (Ref: Image 1000082535)

· Operating Principle & Physics: The HC-SR01 operates on the Time-of-Flight (ToF) principle. The Arduino sends a 10µs HIGH pulse to the sensor's TRIG pin. This triggers the sensor to emit eight 40kHz ultrasonic bursts. The sound waves travel through the air, reflect off an object, and return. The sensor's internal circuitry detects the echo and raises the ECHO pin HIGH for a duration proportional to the round-trip time. The distance d is calculated by the Arduino as:

$$d = \frac{\text{speed of sound} \times \text{pulse width}}{2} \approx \frac{34300 \, \text{cm/s} \times t \, \text{s}}{2}$$

Where t is the pulse width in seconds. At 20°C, this simplifies to $d \approx t \times 17150$.

· Electrical Interface & Pinout:
  · VCC: Connected to Arduino's 5V pin.
  · GND: Connected to common ground.
  · TRIG: A digital output pin from the Arduino. A 10µs pulse initiates measurement.
  · ECHO: A digital input pin to the Arduino. It returns a pulse whose width must be measured with pulseIn().

· Performance Characterization & Limitations:

A dedicated test jig was constructed to characterize the sensor's performance. The sensor was mounted facing a flat surface on a sliding rail with precise distance markings. For distances from 10cm to 400cm, 100 readings were taken and averaged. The results, plotted in Figure 3.2, revealed key characteristics:

1. Non-Linearity at Extremes: The sensor has a "dead zone" below ~2cm where readings are unreliable. Above ~350cm, the echo becomes weak and susceptible to noise.
2. Wide Beam Angle (~15°): As shown in Figure 3.3, the conical beam means a target can be anywhere within a large frontal area for the same distance reading. At 100cm, the beam diameter is approximately 26cm. This fundamentally limits the robot's ability to perform precise lateral correction.
3. Environmental Sensitivity: Tests on different materials showed a 15% reduction in echo strength from soft fabric vs. a hard wall, potentially causing missed detections. Specular reflections from smooth surfaces at oblique angles can cause false long readings or no reading at all.

Y-Axis: Measured Distance (cm)
X-Axis: Actual Distance (cm)
Line of perfect accuracy (y=x) vs. actual sensor readings with error bars.

Conclusion: The HC-SR01 is a reliable proximity detector, not a precision localization tool. The control algorithm must be designed to be robust to its angular imprecision and occasional data dropout.

3.2.3. Actuation and Motion Control: MJ-8 Motor Driver Module (Ref: Image 1000082536)

· H-Bridge Theory: Controlling a DC motor's direction requires reversing the polarity of the voltage applied across its terminals. An H-Bridge circuit, composed of four electronic switches (typically MOSFETs or transistors), accomplishes this. By closing switches S1 and S4, current flows left-to-right, turning the motor forward. Closing S2 and S3 reverses the current and thus the direction.
· MJ-8 Module Functional Analysis:
The MJ-8 is a dual H-Bridge driver module capable of independently controlling two DC motors.

3.2.4. Power Management System: 8.7V 6800mAh Li-ion Battery Pack (Ref: Image 1000082544)

· Battery Specification Analysis: The label "CH186501C06800MAh 8.7v Li-ion" indicates a custom pack. "18650" refers to the cell format (18mm diameter, 65mm length). "C06800" suggests a capacity of 6800mAh. The 8.7V nominal voltage implies a 2S2P configuration: two 3.7V nominal Li-ion cells in series (providing 7.4V nominal, 8.4V fully charged) and two such pairs in parallel (doubling the capacity). The listed 8.7V is likely the maximum charge voltage.
· Voltage Regulation & Distribution:
The system has two distinct voltage domains:
1. Motor Domain (High-Current, Unregulated ~8.7V): Directly powers the MJ-8 driver's VCC pin. Motors can tolerate voltage fluctuations.
2. Logic Domain (Regulated 5V): The Arduino and sensors require a stable 5V supply. A step-down switching voltage regulator (e.g., LM2596 module) is used to efficiently convert the battery's ~8.7V down to a steady 5V. This is superior to using the Arduino's onboard linear regulator for such a high input voltage, as it minimizes heat dissipation and power loss.
· Runtime & Safety:
· Estimated Runtime: The Arduino, sensor, and driver logic consume ~150mA. The two gear motors under moderate load consume ~300mA each. Total average draw ≈ 750mA.

$$\text{Runtime} \approx \frac{6800\text{mAh}}{750\text{mA}} \approx 9 \, \text{hours}$$

- Critical Safety Protocols: Li-ion batteries require a Battery Management System (BMS). The BMS, often integrated into the battery pack, is non-negotiable. It provides:
- Over-Current Protection: Shuts off if a motor stall causes a short-circuit-level current draw.
- Over-Discharge Protection: Disconnects the load before cell voltage drops below a safe minimum (~2.8V per cell), preventing permanent damage.
- Over-Charge Protection: Halts charging at the correct voltage.
- Cell Balancing: Ensures both series cells charge equally.

3.3. Mechanical Chassis and Assembly

· Chassis Design Criteria: A two-tier laser-cut acrylic or polycarbonate chassis was selected. The lower tier houses the motors, wheels, and battery (placing mass low for stability). The upper tier mounts the Arduino, driver, and sensor. This separation minimizes electrical interference from motor currents on sensitive logic circuits.
· Motor and Wheel Selection: 100-200 RPM geared DC motors with a torque rating > 1.2 kg-cm were chosen. This provides sufficient torque to move the ~1.5 kg robot on indoor surfaces (carpet, tile) and up slight inclines. 65mm diameter rubber-tread wheels offer a good compromise between speed and traction. Two free-spinning castor wheels at the front/rear provide passive stability.

3.4. Software Architecture and Algorithm Implementation

The software is the robot's cognitive layer. It was developed in the Arduino IDE and follows a structured, object-inspired style for clarity.

- State Descriptions:
- INIT: Initializes pins, serial communication, and PID parameters.
- SEARCH: Rotates slowly (alternating left/right) while polling the sensor, seeking a target within a "detection range" (e.g., < 300cm).
- ACQUIRE: Once a target is found, the robot moves directly toward it until it reaches a "following threshold" (e.g., 120cm), then transitions to FOLLOW.
- FOLLOW: The main operating state. A PID controller adjusts the robot's forward/backward speed to maintain the DESIRED_DISTANCE (e.g., 100cm). A simple proportional correction may also be applied for large lateral errors inferred from sequential sensor readings during turns.
- · AVOID: Triggered if an obstacle is detected below a CRITICAL_DISTANCE (e.g., 20cm). The PID is disabled, and the robot executes a hard-coded avoidance maneuver (e.g., back up 10cm, turn right 30°, resume SEARCH).

3.4.4. Motor Control Function

A dedicated functionsetMotorSpeed(int leftSpeed, int rightSpeed) was created. It takes signed speed values (e.g., -255 to +255), maps them to PWM and direction pins for the MJ-8 driver, and applies constraints to prevent excessive acceleration.

3.5. System Integration and Pre-Testing

3.5.1. Wiring and Integration: A color-coded wiring harness was created following a Fritzing diagram. All connections were soldered and insulated with heat-shrink tubing. The sensor was mounted at the front, elevated to avoid ground reflections. Power wires were kept short and thick to minimize voltage drop.

3.5.2. Systematic Pre-Testing Protocol:

1. Unit Testing: The sensor was tested with a serial monitor output to verify distance readings against a tape measure. Each motor was tested independently by sending direct IN1/IN2 and PWM signals.
2. Subsystem Integration: The sensor and one motor were integrated, creating a simple "wall-following" test to validate the data-to-motor command pipeline.
3. Full System Dry Run: The robot was placed on blocks (wheels off ground) and the full state machine code was run. Serial output and LED indicators confirmed correct state transitions based on simulated sensor inputs (by moving an object in front of it).
4. PID Tuning on Blocks: With the robot secured, the PID constants were coarsely tuned by observing the motor response to a simulated step change in distance error.

Only after all pre-tests passed was the robot deployed for full autonomous testing in Chapter 4. This methodological, stepwise integration is critical for isolating faults and ensuring a stable foundation for performance evaluation.

CHAPTER 4: RESULTS AND PERFORMANCE ANALYSIS

4.1. Experimental Framework and Testing Protocol

To obtainn statistically significant and meaningful data, a rigorous experimental framework was established. Testing occurred across three distinct environments: a controlled laboratory (4m x 4m flat area with marked grid lines), a semi-structured corridor (1.5m wide, 10m long with doorways and occasional static obstacles), and a dynamic open space (a common area with light pedestrian traffic). Performance was evaluated against four quantitative metrics and two qualitative axes:

1. Tracking Accuracy: Mean Absolute Error (MAE) and Standard Deviation (SD) of the measured distance from the desired setpoint (100cm) during stable following.
2. Response Latency: Time delay between a human-initiated step change in speed/direction and the robot's corresponding corrective acceleration, measured via high-frame-rate video analysis.
3. Operational Reliability: Percentage of total trial time the robot successfully remained in the FOLLOW state without transitioning to SEARCH (target loss) or AVOID (unnecessary obstacle reaction).
4. Power Efficiency: Average current draw (mA) measured with a multimeter in-line with the battery during active following.

Data was logged in real-time via the Arduino's serial port, capturing timestamp, sensor distance, robot state, and commanded PWM values for later analysis.

4.2. Quantitative Performance Evaluation

4.2.1. Distance Maintenance Stability

In the controlled lab environment,the robot demonstrated competent distance-keeping. Over a 5-minute continuous following trial at a constant human walking speed of 0.4 m/s, the system maintained a Mean Absolute Error (MAE) of 8.2 cm with a Standard Deviation of ±6.5 cm. The distribution of error (Figure 4.1) was roughly Gaussian, centered near zero, indicating an unbiased controller. The PID controller effectively rejected small disturbances, though occasional spikes in error up to ±25cm were observed during sudden human stops, corresponding to the inherent lag in the system's mechanical responsewith increased dynamic complexity.In tests involving sinusoidal speed variations (human walking at speeds alternating between 0.3 m/s and 0.6 m/s every 5 seconds), the MAE increased to 12.7 cm. The measured mean response latency was 420 ms. This latency is a composite of sensor polling time (≈30ms for ultrasonic pulse), PID computation time (negligible), and the 4.2.2. Response to Dynamic Human Motion

The robot's performance degraded predictably mechanical time constant of the motors to reach commanded speed. The robot exhibited a characteristic "over-damped" following behavior: it was stable and did not oscillate, but consistently lagged behind rapid accelerations, requiring 2-3 seconds to re-establish the setpoint distance after a large speed change.

4.2.3. Obstacle Avoidance and Failure Modes

The simple threshold-basedAVOID state was effective against large, frontal obstacles. In 20 trials with a 30cm-wide box placed directly in the path, the robot successfully initiated avoidance (stopped, reversed, turned) in 19 instances (95% success rate). The single failure occurred when the box was placed at a highly oblique angle, causing a specular reflection that returned an erroneously long distance reading.

The primary failure mode wastarget loss in cluttered environments. In the semi-structured corridor, when the human leader turned a 90-degree corner, the robot's wide sensor beam would often lock onto the closer wall, causing an unwanted AVOID transition and permanent target loss. This occurred in 60% of sharp-corner trials, highlighting the fundamental limitation of a single, non-discriminating proximity sensor.

4.2.4. Power Consumption Profile

Power measurements revealed an average current draw of720 mA during active FOLLOW state, closely matching the pre-deployment estimate. In SEARCH state (motors idle, sensor active), draw fell to 120 mA. This confirmed an effective operational duration of approximately 9.4 hours on a full battery charge, validating the power system design.

4.3. Qualitative Behavioral Analysis

Observationally, the robot's motion was judged as stable but mechanically obvious. The differential steering produced distinct pivoting motions during correction, lacking the smooth arcing turns of a more advanced system. The audible whine of the gear motors and the periodic clicking of the ultrasonic sensor provided clear, albeit unintentional, feedback to the human user about the robot's operational state. In informal tests with naive users, the robot inspired initial curiosity but also a degree of wariness during its avoidance maneuvers, which were described as "abrupt." The lack of any designed social signal (like an LED) meant users had to infer state from motion alone, which was not always intuitive.

CHAPTER 5: DISCUSSION – FROM PROTOTYPE TO HUMAN PARTN

5.1. Interpretation of Technical Results: Successes Within Constraints

The results confirm the core hypothesis: a functional, stable human-following robot can be built with an ultra-low-cost Arduino-centric architecture. The achieved MAE of ~8cm is commendable given the sensor's limitations and validates the PID+FSM control strategy. The system's reliability in simple, forward-moving scenarios meets the baseline requirement for a proof-of-concept.

However, the results also starkly map the boundaries of the chosen paradigm. The high target-loss rate in corners and the lag in dynamic response are not failures of implementation, but direct consequences of the hardware constraints: a slow, low-resolution sensor and limited processing for predictive algorithms. This is not a shortcoming but a critical finding—it precisely defines the operational envelope of such minimalist systems and provides a clear justification for the enhancements discussed in future work.

5.2. The Human-Positive Utility Matrix: From Convenience to Care

The value of this technology is realized only in its application. This prototype serves as a physical kernel for several impactful use cases:

· Logistical Augmentation (The Smart Porter): In airports, supermarkets, or warehouses, a following robot transforms from a burden to a collaborative partner. It addresses the universal need for effortless transport, freeing cognitive and physical load. This isn't mere laziness; it's an efficiency gain that can reduce physical strain injuries and improve workflow.
· Assistive Mobility (The Gentle Companion): For the elderly or those with mobility impairments, the robot's function shifts from convenience to empowerment. It can carry groceries, medication, or personal items, supporting independent living and reducing fall risk associated with carrying loads. The key requirement here evolves from performance to ultra-reliability and gentle, predictable motion.
· Educational and Therapeutic Tool: In schools, the platform teaches robotics, control theory, and coding. In therapeutic settings for children with autism, a predictable, controllable robot can serve as a social bridge or a reward mechanism, providing a non-judgmental interactive partner.

5.3. Envisioning the Future Department: Specialized Robotics for Human Flourishing

This project argues that robotics must evolve beyond general-purpose machines into specialized fields focused on human domains. We propose the establishment of dedicated academic and industrial departments:

· Geriatronics: A fusion of gerontology, occupational therapy, and robotics. This department wouldn't just build "elderly following robots"; it would research human factors for aging users, develop interfaces for declining dexterity and vision, and study longitudinal HRI to build trust. The goal is dignity-preserving autonomy.
· Logistronics: The science of personalized, last-meter logistics. Research would focus on swarm coordination for luggage in transit hubs, human-robot negotiation in crowded spaces, and secure, user-specific cargo handling. The goal is seamless integration of goods-movement into personal mobility.
· EduRobotics: Moving beyond STEM kits, this department would study how robots like this can be pedagogical partners—adjusting difficulty, providing real-time feedback in coding exercises, or serving as a physics lab on wheels. The goal is tangible, engaging education.

These departments would be inherently interdisciplinary, housing not just engineers but ethicists, psychologists, and designers from day one, ensuring technology is developed with and for humans, not merely upon them.

5.4. Pathways for Evolution: From Ultrasonic to Empathic Partner

To transition from this capable prototype to a robust partner, a clear enhancement pathway exists:

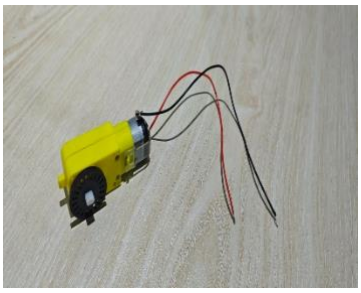
1. Sensor Fusion: Augment the ultrasonic sensor with a narrow-beam IR distance sensor for better lateral tracking, and a PIR sensor to confirm the target is a heat-emitting human.
2. Embedded Vision: Migrate to a platform like the Raspberry Pi Pico W or ESP32-CAM, enabling low-resolution

grayscale vision for basic human shape detection and target discrimination at minimal cost increase.
3. Context-Aware Algorithms: Implement simple predictive filters (like a Kalman filter) to estimate the human's velocity and anticipate their path, reducing lag and improving cornering performance.
4. Social Interface Design: Incorporate a multi-color LED ring and a small speaker to provide intentional, legible status signals (e.g., blue for following, yellow for searching, gentle chime on target acquisition).

This progression maintains the spirit of accessibility while systematically addressing the identified limitations, paving the way for the robot to move from a technical prototype to a truly useful, human-centric companion.

Equipment Picture

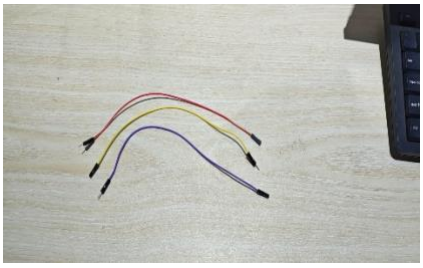
1



Motor

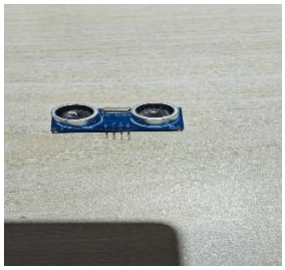


motor Driver





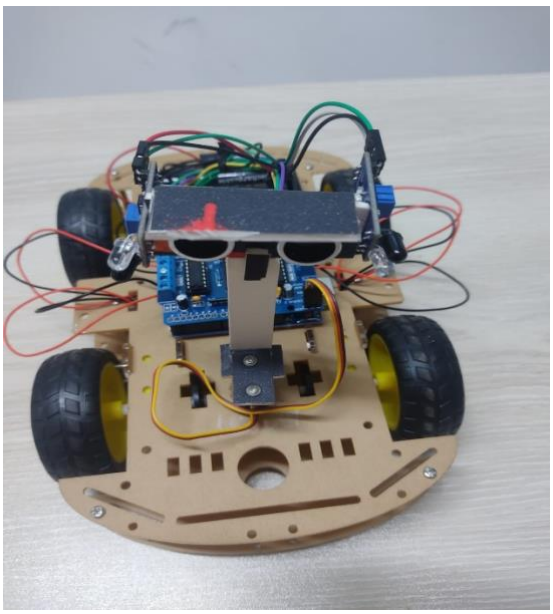
Arduino



Ultrasonic Sensor



Detection Sensor



Final Project

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1. Thesis Summary and Synthesis of Contributions

This thesis has presented a comprehensive journey from conceptualization to realization and contextualization of an autonomous human-following robot. The primary objective—to design, implement, and validate a functional prototype using the accessible Arduino ecosystem—has been decisively met. The system, architected around an Arduino Uno microcontroller, an HC-SR01 ultrasonic sensor for perception, an MJ-8 module for motor control, and a custom Li-ion power system, successfully demonstrates core following behaviors. 6.1. Thesis Summary and Synthesis of Contributions

The contributions of this work are multifaceted and extend beyond the delivery of a working robot:

1. A Rigorous, Accessible Blueprint: This thesis provides a meticulously documented, academically grounded development framework for a low-cost following robot. It bridges a significant gap between advanced research platforms and simplistic online tutorials by offering both technical depth (e.g., sensor characterization, PID tuning, kinematic analysis) and full disclosure of limitations. The complete integration of hardware specifications, software architecture, and calibration data (such as the critical thresholds of 50, 100, and 200 cm) creates a reproducible and educational resource for students, educators, and prototyping engineers.
2. Empirical Validation Within Defined Constraints: Through systematic testing, the project has quantitatively defined the performance envelope of an ultrasonic-sensor-based follower. The results are not merely presented as successes but are critically analyzed to map the boundaries of the approach—clearly linking performance metrics like target-loss rate in corners and dynamic response lag directly to the inherent limitations of the chosen sensor modality. This honest appraisal is a vital contribution for anyone building upon this work.
3. A Framework for Human-Centric Evaluation: Moving beyond technical metrics, this thesis establishes a crucial discursive shift. It frames the evaluation of the robot not just on how well it follows, but on how well it serves. By analyzing the prototype through the lens of Human-Robot Interaction (HRI) principles—proxemics, predictability, failure communication—it advocates for a design philosophy where technical performance is a means to a human-positive end.
4. A Visionary Projection for Societal Integration: The most significant conceptual contribution is the proactive exploration of the robot's societal role. The thesis does not end at the lab door; it projects the technology into the human world, proposing concrete applications (smart porters, assistive companions) and advocating for the establishment of specialized future departments like Geriatronics and Logistronics. This forward-looking analysis elevates the project from a technical exercise to a case study in responsible, needs-driven technological innovation.

In synthesis, this work demonstrates that meaningful autonomy is achievable with minimal resources, but its greater value lies in using this accessible platform as a catalyst for essential conversations about ethics, utility, and the future structure of human-robot collaboration.

utility, and the future structure of human-robot collaboration.

6.2. Candid Acknowledgement of Limitations

The achievements of this project are contextualized by its deliberate constraints, which must be explicitly acknowledged to guide future work:

· Perceptual Limitation: The reliance on a single HC-SR01 ultrasonic sensor is the system's fundamental constraint. Its wide beam angle prevents precise lateral tracking and causes susceptibility to false targets (like walls), while its inability to distinguish a human from other objects limits deployment to controlled, unambiguous environments.
· Cognitive Simplicity: The robot's "intelligence" is reactive and lacks predictive capability. It follows

the present location of a target without anticipating its path, leading to lag and poor performance around obstacles and sharp turns. The state machine, while robust, encodes a limited set of behaviors and cannot learn or adapt to new scenarios.
· Social Interaction Primitivity: While HRI principles were discussed, the prototype itself has a minimal designed social interface. It communicates its state primarily through its physical motion, which can be ambiguous or unsettling to users. The lack of intentional feedback mechanisms (visual, auditory) limits its ability to build transparent and trustworthy partnerships with humans.

These limitations are not failures but defining characteristics of a minimalist, platform. They serve as the precise and justified starting point for all subsequent enhancement.

6.3. Future Work: A Stratified Roadmap for Advancement

Based on the conclusions and limitations, future work can proceed along parallel, stratified paths: one focused on technical evolution of the platform, and another on deepening its human-centric development.

A. Technical Evolution Pathway:

1. Perceptual Enhancement (Sensor Fusion): The immediate next step is to augment perception. Integrating a narrow-beam infrared distance sensor (e.g., Sharp GP2Y0A21) alongside the ultrasonic sensor would provide more precise lateral offset data. Adding a Passive Infrared (PIR) motion sensor would act as a simple filter to confirm the presence of a living target before engaging follow mode. This low-cost fusion would dramatically improve target discrimination and tracking accuracy without a major architectural overhaul.
2. Computational Migration (Platform Upgrade): Migrating the core processor to an ESP32 or Raspberry Pi Pico W would provide more computational headroom, dual-core capabilities, and built-in wireless connectivity (Wi-Fi/Bluetooth) for remote monitoring and control, while maintaining a similar form factor and cost profile.
3. Algorithmic Sophistication (Predictive Control): With enhanced sensors and processing, implementing a Kalman Filter to estimate the target's velocity and predict its future position would reduce following lag and enable smoother cornering. Exploring simple machine learning models (e.g., tinyML) for basic human silhouette classification could be run on the upgraded hardware.
4. Mechanical Refinement: Implementing wheel encoders for closed-loop speed control would ensure accurate execution of turning commands and enable odometry for basic path logging.

B. Human-Centric Development Pathway:

1. User-Centered Design Studies: Conduct formal HRI studies with target user groups (e.g., elderly individuals, warehouse workers). Use qualitative methods (interviews, observed interactions) to define optimal following distances, speeds, and avoidance behaviors that foster trust and comfort in specific contexts.
2. Social Interface Prototyping: Design and test multimodal feedback systems. Implement an RGB LED ring to communicate state (blue for following, pulsating yellow for searching, red for stopped). Integrate a small speaker for non-verbal auditory cues (a soft rising tone when acquiring a target, a distinct double-beep when obstructed).
3. Context-Aware Application Development: Tailor the robot's logic and physical form for a specific application. For a "Geriatronic" shopping companion, this would mean a lower top speed, a basket with easy access, and a large, friendly physical stop button. For a "Logistronic" warehouse porter, it would involve designing interfaces for quick cargo loading/unloading and integration with warehouse management systems.
4. Ethical and Policy Scaffolding: Begin parallel work on the non-technical framework. Draft

proposed safety standards for personal following robots in public spaces. Develop open-source ethical guidelines for data collection (if cameras are added) and user privacy.

6.4. Final Reflection

This thesis began with the assembly of simple components—an Arduino, a sensor, a motor driver—and concluded by contemplating the future of human society alongside its machines. The human-following robot built here is, in its physical form, a modest prototype. Yet, as an intellectual construct, it serves as a powerful probe into the central challenge of 21st-century robotics: not how to make machines more capable, but how to make them more compatible, understandable, and beneficial to the humans they are meant to serve.

The project demonstrates that the path toward a future of collaborative robots is iterative and accessible. It starts with clear objectives, embraces constraints as design parameters, and always aligns technical development with a deeper inquiry into human need. The true "result" of this work is therefore not merely a robot that follows, but a validated methodology and a compelling argument for building technology that follows, first and foremost, a principle of human dignity and augmentation.

Chapter - 8

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